

Objective

Build business reports using the Oracle HR Schema while demonstrating a strong understanding of table relationships, JOIN operations, and enterprise reporting requirements.

Scope

1. HR Database Exploration

- **EMPLOYEES: (Self-Reference):** One-to-Many due to `manager_id` (An employee reports to another employee).
- **DEPARTMENTS:** Has a relationship to Locations via `location_id`.
- **JOBS:** Has a relationship to employees via `job_id` (An employee holds one job role).
- **LOCATIONS:** Has a relationship to Countries via `country_id`.
- **COUNTRIES:** Has a relationship to locations via `country_id`.
- **REGIONS:** Has a relationship to COUNTRIES via `region_id`.

2. Employee & Department Reports

Create SQL queries for:

1. Employee ID, full name, and department name (employees with departments only).

```
SELECT e.employee_id, e.first_name || ' ' || e.last_name AS employee_name,  
       d.department_name FROM employees e INNER JOIN departments d ON  
       e.department_id = d.department_id;
```

SQL All Rows Fetched: 106 in 0.018 seconds		
EMPLOYEE_ID	EMPLOYEE_NAME	DEPARTMENT_NAME
1	200 Jennifer Whalen	Administration
2	201 Michael Hartstein	Marketing
3	202 Pat Fay	Marketing
4	114 Den Raphaely	Purchasing
5	119 Karen Colmenares	Purchasing
6	115 Alexander Khoo	Purchasing
7	116 Shelli Baida	Purchasing
8	117 Sigal Tobias	Purchasing
9	118 Guy Himuro	Purchasing
10	203 Susan Mavris	Human Resources
11	198 Donald OConnell	Shipping
12	199 Douglas Grant	Shipping
13	120 Matthew Weiss	Shipping
14	121 Adam Fripp	Shipping
15	122 Payam Kaufling	Shipping
16	123 Shanta Vollman	Shipping
17	124 Kevin Mourgous	Shipping
18	125 Julia Nayer	Shipping
19	126 Irene Mikkilineni	Shipping
20	127 James Landry	Shipping
21	128 Steven Markle	Shipping
22	129 Laura Bissot	Shipping
23	130 Mozhe Atkinson	Shipping

- Employee full name, job title, and salary (employees with valid jobs only).

```
SELECT e.first_name || ' ' || e.last_name AS employee_name, j.job_title,
TO_CHAR(e.salary, '$99,999.00') AS salary FROM employees e INNER JOIN jobs j
ON e.job_id = j.job_id;
```

Query Result x			
SQL Fetched 50 rows in 0.007 seconds			
EMPLOYEE_NAME	JOB_TITLE	SALARY	
1 William Gietz	Public Accountant	\$8,300.00	
2 Shelley Higgins	Accounting Manager	\$12,008.00	
3 Jennifer Whalen	Administration Assistant	\$4,400.00	
4 Steven King	President	\$24,000.00	
5 Neena Kochhar	Administration Vice President	\$17,000.00	
6 Lex De Haan	Administration Vice President	\$17,000.00	
7 Ismael Sciarra	Accountant	\$7,700.00	
8 John Chen	Accountant	\$8,200.00	
9 Luis Popp	Accountant	\$6,900.00	
10 Jose Manuel Urman	Accountant	\$7,800.00	
11 Daniel Faviet	Accountant	\$9,000.00	
12 Nancy Greenberg	Finance Manager	\$12,008.00	
13 Susan Mavris	Human Resources Representative	\$6,500.00	
14 Bruce Ernst	Programmer	\$6,000.00	
15 David Austin	Programmer	\$4,800.00	
16 Valli Pataballa	Programmer	\$4,800.00	
17 Diana Lorentz	Programmer	\$4,200.00	
18 Alexander Hunold	Programmer	\$9,000.00	
19 Michael Hartstein	Marketing Manager	\$13,000.00	
20 Pat Fay	Marketing Representative	\$6,000.00	
21 Hermann Baer	Public Relations Representative	\$10,000.00	
22 Karen Colmenares	Purchasing Clerk	\$2,500.00	
23 Alexander Khoo	Purchasing Clerk	\$3,100.00	

3. Employee, department, and department city.

SELECT e.first_name || ' ' || e.last_name AS employee_name, d.department_name, l.city FROM employees e INNER JOIN departments d ON e.department_id = d.department_id INNER JOIN locations l ON d.location_id = l.location_id;

Query Result x			
SQL Fetched 50 rows in 0.012 seconds			
EMPLOYEE_NAME	DEPARTMENT_NAME	CITY	
1 Ellen Abel	Sales	Oxford	
2 Sunder Ande	Sales	Oxford	
3 Mozhe Atkinson	Shipping	South San Francisco	
4 David Austin	IT	Southlake	
5 Hermann Baer	Public Relations	Munich	
6 Shelli Baida	Purchasing	Seattle	
7 Amit Banda	Sales	Oxford	
8 Elizabeth Bates	Sales	Oxford	
9 Sarah Bell	Shipping	South San Francisco	
10 David Bernstein	Sales	Oxford	
11 Laura Bissot	Shipping	South San Francisco	
12 Harrison Bloom	Sales	Oxford	
13 Alexis Bull	Shipping	South San Francisco	
14 Anthony Cabrio	Shipping	South San Francisco	
15 Gerald Cambrault	Sales	Oxford	
16 Nanette Cambrault	Sales	Oxford	
17 John Chen	Finance	Seattle	
18 Kelly Chung	Shipping	South San Francisco	
19 Karen Colmenares	Purchasing	Seattle	
20 Curtis Davies	Shipping	South San Francisco	
21 Lex De Haan	Executive	Seattle	
22 Julia Dellinger	Shipping	South San Francisco	
23 Jennifer Dilly	Shipping	South San Francisco	

4. Employee, department, country, and region.

SELECT e.first_name || ' ' || e.last_name AS employee_name, d.department_name, c.country_name, r.region_name FROM employees e INNER JOIN departments d ON e.department_id = d.department_id INNER JOIN locations l ON d.location_id =

l.location_id INNER JOIN countries c ON l.country_id = c.country_id INNER JOIN regions r ON c.region_id = r.region_id;

 SQL | Fetched 50 rows in 0.019 seconds

	EMPLOYEE_NAME	DEPARTMENT_NAME	COUNTRY_NAME	REGION_NAME
1	Ellen Abel	Sales	United Kingdom	Europe
2	Sundar Ande	Sales	United Kingdom	Europe
3	Mozhe Atkinson	Shipping	United States of America	Americas
4	David Austin	IT	United States of America	Americas
5	Hermann Baer	Public Relations	Germany	Europe
6	Shelli Baida	Purchasing	United States of America	Americas
7	Amit Banda	Sales	United Kingdom	Europe
8	Elizabeth Bates	Sales	United Kingdom	Europe
9	Sarah Bell	Shipping	United States of America	Americas
10	David Bernstein	Sales	United Kingdom	Europe
11	Laura Bissot	Shipping	United States of America	Americas
12	Harrison Bloom	Sales	United Kingdom	Europe
13	Alexis Bull	Shipping	United States of America	Americas
14	Anthony Cabrio	Shipping	United States of America	Americas
15	Gerald Cambrault	Sales	United Kingdom	Europe
16	Nanette Cambrault	Sales	United Kingdom	Europe
17	John Chen	Finance	United States of America	Americas
18	Kelly Chung	Shipping	United States of America	Americas
19	Karen Colmenares	Purchasing	United States of America	Americas
20	Curtis Davies	Shipping	United States of America	Americas
21	Lex De Haan	Executive	United States of America	Americas
22	Julia Dellinger	Shipping	United States of America	Americas
23	Jennifer Dilly	Shipping	United States of America	Americas

3. Organization Structure Analysis

Create reports for:

5. Complete employee list including employees without departments.

```
SELECT e.employee_id, e.first_name || ' ' || e.last_name AS employee_name,
NVL(d.department_name, 'Unassigned / Bench') AS department_name FROM employees e
LEFT OUTER JOIN departments d ON e.department_id = d.department_id;
```

	EMPLOYEE_ID	EMPLOYEE_NAME	DEPARTMENT_NAME
1	200	Jennifer Whalen	Administration
2	201	Michael Hartstein	Marketing
3	202	Pat Fay	Marketing
4	114	Den Raphaely	Purchasing
5	115	Alexander Khoo	Purchasing
6	116	Shelli Baida	Purchasing
7	117	Sigal Tobias	Purchasing
8	118	Guy Himuro	Purchasing
9	119	Karen Colmenares	Purchasing
10	203	Susan Mavris	Human Resources
11	198	Donald OConnell	Shipping
12	199	Douglas Grant	Shipping
13	120	Matthew Weiss	Shipping
14	121	Adam Fripp	Shipping
15	122	Payam Kaufling	Shipping
16	123	Shanta Vollman	Shipping
17	124	Kevin Mourgous	Shipping
18	125	Julia Nayer	Shipping
19	126	Irene Mikkilineni	Shipping
20	127	James Landry	Shipping
21	128	Steven Markle	Shipping
22	129	Laura Bissot	Shipping
23	130	Mozhe Atkinson	Shipping

6. Complete department list including departments without employees.

```
SELECT d.department_id, d.department_name, e.first_name || ' ' || e.last_name AS
employee_name FROM employees e RIGHT OUTER JOIN departments d ON
e.department_id = d.department_id;
```

SQL | Fetched 50 rows in 0.005 seconds

DEPARTMENT_ID	DEPARTMENT_NAME	EMPLOYEE_NAME
1	10 Administration	Jennifer Whalen
2	20 Marketing	Pat Fay
3	20 Marketing	Michael Hartstein
4	30 Purchasing	Sigal Tobias
5	30 Purchasing	Karen Colmenares
6	30 Purchasing	Shelli Baida
7	30 Purchasing	Den Raphaely
8	30 Purchasing	Alexander Khoo
9	30 Purchasing	Guy Himuro
10	40 Human Resources	Susan Mavris
11	50 Shipping	Kevin Feeney
12	50 Shipping	Jean Fleaur
13	50 Shipping	Adam Fripp
14	50 Shipping	Timothy Gates
15	50 Shipping	Ki Gee
16	50 Shipping	Girard Geoni
17	50 Shipping	Douglas Grant
18	50 Shipping	Vance Jones
19	50 Shipping	Payam Kaufling
20	50 Shipping	Renske Ladwig
21	50 Shipping	Britney Everett
22	50 Shipping	Jennifer Dilly
23	50 Shipping	Julia Dellinger

7. Departments that currently have no employees assigned.

```
SELECT d.department_id, d.department_name FROM departments d LEFT OUTER JOIN
employees e ON d.department_id = e.department_id WHERE e.employee_id IS NULL;
```

Query Result x

SQL | All Rows Fetched: 16 in 0.006 seconds

DEPARTMENT_ID	DEPARTMENT_NAME
1	120 Treasury
2	130 Corporate Tax
3	140 Control And Credit
4	150 Shareholder Services
5	160 Benefits
6	170 Manufacturing
7	180 Construction
8	190 Contracting
9	200 Operations
10	210 IT Support
11	220 NOC
12	230 IT Helpdesk
13	240 Government Sales
14	250 Retail Sales
15	260 Recruiting
16	270 Payroll

4. Management Hierarchy Analysis

Create reports for:

8. Employee name and direct manager name.

```
SELECT e.first_name || ' ' || e.last_name AS employee_name, m.first_name || ' ' ||  
m.last_name AS manager_name FROM employees e INNER JOIN employees m ON  
e.manager_id = m.employee_id;
```

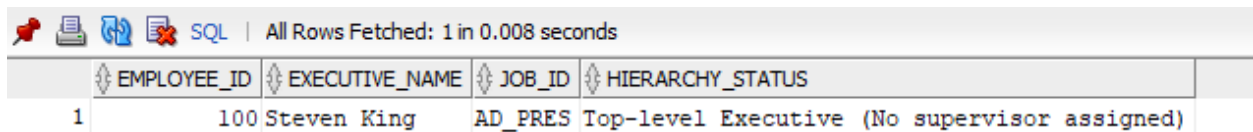


SQL | Fetched 100 rows in 0.014 seconds

	EMPLOYEE_NAME	MANAGER_NAME
1	Gerald Cambrault	Steven King
2	Lex De Haan	Steven King
3	Alberto Errazuriz	Steven King
4	Adam Fripp	Steven King
5	Michael Hartstein	Steven King
6	Payam Kaufling	Steven King
7	Neena Kochhar	Steven King
8	Kevin Mourgous	Steven King
9	Karen Partners	Steven King
10	Den Raphaely	Steven King
11	John Russell	Steven King
12	Shanta Vollman	Steven King
13	Matthew Weiss	Steven King
14	Eleni Zlotkey	Steven King
15	Hermann Baer	Neena Kochhar
16	Nancy Greenberg	Neena Kochhar

9. Employees without managers and provide a brief explanation.

```
SELECT e.employee_id, e.first_name || ' ' || e.last_name AS executive_name, e.job_id, 'Top-  
level Executive (No supervisor assigned)' AS hierarchy_status FROM employees e WHERE  
e.manager_id IS NULL;
```



SQL | All Rows Fetched: 1 in 0.008 seconds

	EMPLOYEE_ID	EXECUTIVE_NAME	JOB_ID	HIERARCHY_STATUS
1	100	Steven King	AD_PRES	Top-level Executive (No supervisor assigned)

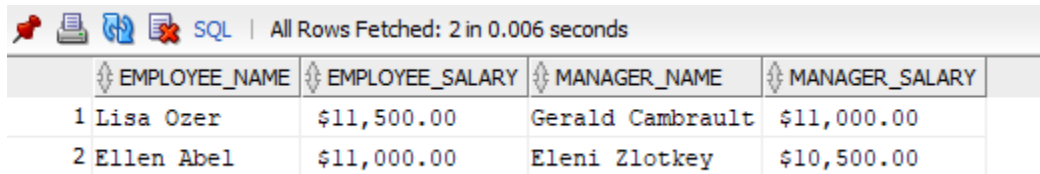
Based on the output, it returns one row which means that the Executive_name : Steven king is the top level executive that has no supervisor.

5. Advanced Business Analysis

Create reports for:

10. Employees earning more than their managers.

```
SELECT e.first_name || ' ' || e.last_name AS employee_name, TO_CHAR(e.salary, '$99,999.00') AS employee_salary, m.first_name || ' ' || m.last_name AS manager_name, TO_CHAR(m.salary, '$99,999.00') AS manager_salary FROM employees e INNER JOIN employees m ON e.manager_id = m.employee_id WHERE e.salary > m.salary;
```

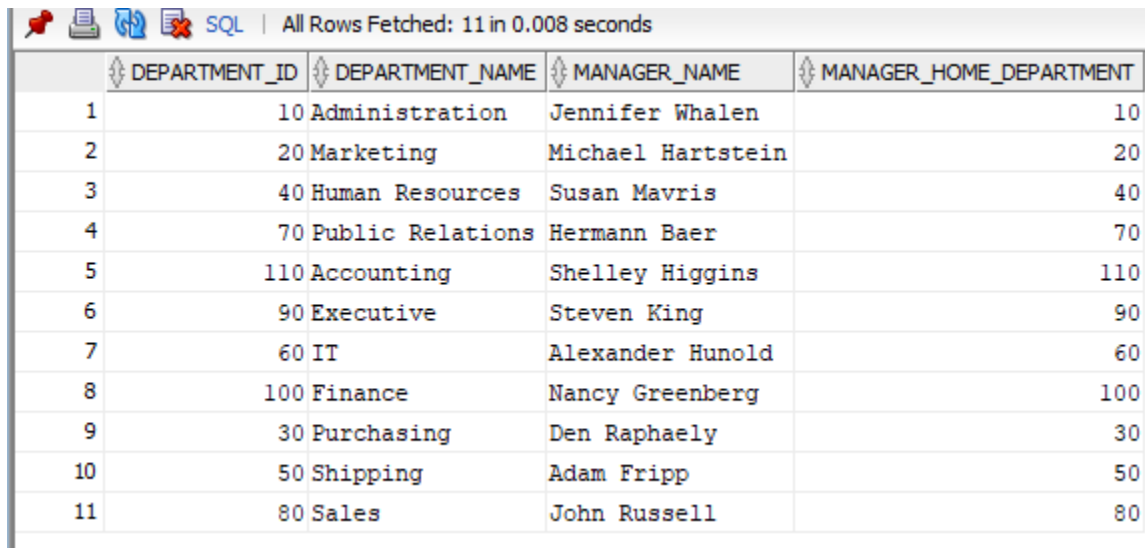


SQL | All Rows Fetched: 2 in 0.006 seconds

	EMPLOYEE_NAME	EMPLOYEE_SALARY	MANAGER_NAME	MANAGER_SALARY
1	Lisa Ozer	\$11,500.00	Gerald Cambrault	\$11,000.00
2	Ellen Abel	\$11,000.00	Eleni Zlotkey	\$10,500.00

11. Departments where the department manager works within the same department.

```
SELECT d.department_id, d.department_name, e.first_name || ' ' || e.last_name AS manager_name, e.department_id AS manager_home_department FROM departments d INNER JOIN employees e ON d.manager_id = e.employee_id WHERE e.department_id = d.department_id;
```



SQL | All Rows Fetched: 11 in 0.008 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_NAME	MANAGER_HOME_DEPARTMENT
1	10	Administration	Jennifer Whalen	10
2	20	Marketing	Michael Hartstein	20
3	40	Human Resources	Susan Mavris	40
4	70	Public Relations	Hermann Baer	70
5	110	Accounting	Shelley Higgins	110
6	90	Executive	Steven King	90
7	60	IT	Alexander Hunold	60
8	100	Finance	Nancy Greenberg	100
9	30	Purchasing	Den Raphaely	30
10	50	Shipping	Adam Fripp	50
11	80	Sales	John Russell	80

6. Executive Dashboard Reports

Create professional reports for:

- Employee, job title, department, city, and country.

```
SELECT e.employee_id, e.first_name || ' ' || e.last_name AS employee_name,  
j.job_title, NVL(d.department_name, 'No Department Assigned') AS  
department_name, l.city, c.country_name FROM employees e INNER JOIN jobs j  
ON e.job_id = j.job_id LEFT JOIN departments d ON e.department_id =  
d.department_id LEFT JOIN locations l ON d.location_id = l.location_id LEFT JOIN  
countries c ON l.country_id = c.country_id;
```

SQL | Fetched 100 rows in 0.012 seconds

	EMPLOYEE_ID	EMPLOYEE_NAME	JOB_TITLE	DEPARTMENT_NAME	CITY	COUNTRY_NAME
1	200	Jennifer Whalen	Administration Assistant	Administration	Seattle	United States of America
2	201	Michael Hartstein	Marketing Manager	Marketing	Toronto	Canada
3	202	Pat Fay	Marketing Representative	Marketing	Toronto	Canada
4	119	Karen Colmenares	Purchasing Clerk	Purchasing	Seattle	United States of America
5	115	Alexander Khoo	Purchasing Clerk	Purchasing	Seattle	United States of America
6	116	Shelli Baida	Purchasing Clerk	Purchasing	Seattle	United States of America
7	117	Sigal Tobias	Purchasing Clerk	Purchasing	Seattle	United States of America
8	118	Guy Himuro	Purchasing Clerk	Purchasing	Seattle	United States of America
9	114	Den Raphaely	Purchasing Manager	Purchasing	Seattle	United States of America
10	203	Susan Mavris	Human Resources Representative	Human Resources	London	United Kingdom
11	198	Donald OConnell	Shipping Clerk	Shipping	South San Francisco	United States of America
12	199	Douglas Grant	Shipping Clerk	Shipping	South San Francisco	United States of America
13	180	Winston Taylor	Shipping Clerk	Shipping	South San Francisco	United States of America
14	181	Jean Fleaur	Shipping Clerk	Shipping	South San Francisco	United States of America
15	182	Martha Sullivan	Shipping Clerk	Shipping	South San Francisco	United States of America
16	183	Girard Geoni	Shipping Clerk	Shipping	South San Francisco	United States of America
17	184	Nandita Sarchand	Shipping Clerk	Shipping	South San Francisco	United States of America
18	185	Alexis Bull	Shipping Clerk	Shipping	South San Francisco	United States of America
19	186	Julia Dellinger	Shipping Clerk	Shipping	South San Francisco	United States of America
20	187	Anthony Cabrio	Shipping Clerk	Shipping	South San Francisco	United States of America
21	188	Kelly Chung	Shipping Clerk	Shipping	South San Francisco	United States of America
22	189	Jennifer Dilly	Shipping Clerk	Shipping	South San Francisco	United States of America
23	190	Timothy Gates	Shipping Clerk	Shipping	South San Francisco	United States of America

- Department, manager, city, and country (including departments without managers).

```
SELECT d.department_name, NVL(m.first_name || ' ' || m.last_name, 'VACANT / NO
MANAGER') AS manager_name, l.city, c.country_name FROM departments d
LEFT OUTER JOIN employees m ON d.manager_id = m.employee_id
LEFT OUTER JOIN locations l ON d.location_id = l.location_id
LEFT OUTER JOIN countries c ON l.country_id = c.country_id;
```

    SQL | All Rows Fetched: 27 in 0.01 seconds

	DEPARTMENT_NAME	MANAGER_NAME	CITY	COUNTRY_NAME
1	Executive	Steven King	Seattle	United States of America
2	IT	Alexander Hunold	Southlake	United States of America
3	Finance	Nancy Greenberg	Seattle	United States of America
4	Purchasing	Den Raphaely	Seattle	United States of America
5	Shipping	Adam Fripp	South San Francisco	United States of America
6	Sales	John Russell	Oxford	United Kingdom
7	Administration	Jennifer Whalen	Seattle	United States of America
8	Marketing	Michael Hartstein	Toronto	Canada
9	Human Resources	Susan Mavris	London	United Kingdom
10	Public Relations	Hermann Baer	Munich	Germany
11	Accounting	Shelley Higgins	Seattle	United States of America
12	Treasury		Seattle	United States of America
13	Corporate Tax		Seattle	United States of America
14	Control And Credit		Seattle	United States of America
15	Shareholder Services		Seattle	United States of America
16	Benefits		Seattle	United States of America
17	Manufacturing		Seattle	United States of America
18	Construction		Seattle	United States of America
19	Contracting		Seattle	United States of America
20	Operations		Seattle	United States of America
21	IT Support		Seattle	United States of America
22	NOC		Seattle	United States of America
23	IT Helpdesk		Seattle	United States of America

- Employees working in the same department as their manager.

```
SELECT e.first_name || ' ' || e.last_name AS employee_name, m.first_name || ' ' ||
m.last_name AS manager_name, d.department_name FROM employees e INNER
JOIN employees m ON e.manager_id = m.employee_id INNER JOIN departments d
ON e.department_id = d.department_id WHERE e.department_id =
m.department_id;
```

 SQL | Fetched 50 rows in 0.034 seconds

	EMPLOYEE_NAME	MANAGER_NAME	DEPARTMENT_NAME
1	Pat Fay	Michael Hartstein	Marketing
2	William Gietz	Shelley Higgins	Accounting
3	Neena Kochhar	Steven King	Executive
4	Lex De Haan	Steven King	Executive
5	Bruce Ernst	Alexander Hunold	IT
6	Diana Lorentz	Alexander Hunold	IT
7	Valli Pataballa	Alexander Hunold	IT
8	David Austin	Alexander Hunold	IT
9	John Chen	Nancy Greenberg	Finance
10	Ismael Sciarra	Nancy Greenberg	Finance
11	Jose Manuel Urman	Nancy Greenberg	Finance
12	Luis Popp	Nancy Greenberg	Finance
13	Daniel Faviet	Nancy Greenberg	Finance
14	Karen Colmenares	Den Raphaely	Purchasing
15	Alexander Khoo	Den Raphaely	Purchasing
16	Shelli Baida	Den Raphaely	Purchasing
17	Sigal Tobias	Den Raphaely	Purchasing
18	Guy Himuro	Den Raphaely	Purchasing
19	Julia Nayer	Matthew Weiss	Shipping
20	Irene Mikkilineni	Matthew Weiss	Shipping
21	James Landry	Matthew Weiss	Shipping
22	Steven Markle	Matthew Weiss	Shipping
23	Winston Taylor	Matthew Weiss	Shipping

- Employees whose managers work in different departments.

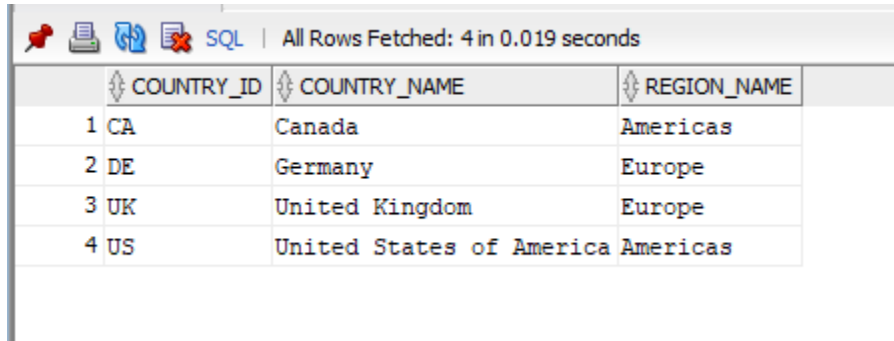
```
SELECT e.first_name || ' ' || e.last_name AS employee_name, ed.department_name
AS employee_department, m.first_name || ' ' || m.last_name AS manager_name,
md.department_name AS manager_department FROM employees e INNER JOIN
employees m ON e.manager_id = m.employee_id INNER JOIN departments ed ON
e.department_id = ed.department_id INNER JOIN departments md ON
m.department_id = md.department_id WHERE e.department_id !=
m.department_id;
```

SQL | All Rows Fetched: 18 in 0.008 seconds

	EMPLOYEE_NAME	EMPLOYEE_DEPARTMENT	MANAGER_NAME	MANAGER_DEPARTMENT
1	Michael Hartstein	Marketing	Steven King	Executive
2	Den Raphaely	Purchasing	Steven King	Executive
3	Matthew Weiss	Shipping	Steven King	Executive
4	Adam Fripp	Shipping	Steven King	Executive
5	Payam Kaufling	Shipping	Steven King	Executive
6	Shanta Vollman	Shipping	Steven King	Executive
7	Kevin Mourgous	Shipping	Steven King	Executive
8	John Russell	Sales	Steven King	Executive
9	Karen Partners	Sales	Steven King	Executive
10	Alberto Errazuriz	Sales	Steven King	Executive
11	Gerald Cambrault	Sales	Steven King	Executive
12	Eleni Zlotkey	Sales	Steven King	Executive
13	Jennifer Whalen	Administration	Neena Kochhar	Executive
14	Susan Mavris	Human Resources	Neena Kochhar	Executive
15	Hermann Baer	Public Relations	Neena Kochhar	Executive
16	Nancy Greenberg	Finance	Neena Kochhar	Executive
17	Shelley Higgins	Accounting	Neena Kochhar	Executive
18	Alexander Hunold	IT	Lex De Haan	Executive

- Distinct countries containing company departments.

```
SELECT DISTINCT c.country_id, c.country_name, r.region_name FROM countries c
INNER JOIN locations l ON c.country_id = l.country_id INNER JOIN departments d
ON l.location_id = d.location_id INNER JOIN regions r ON c.region_id = r.region_id
ORDER BY c.country_name;
```



The image shows a screenshot of a database query results window. At the top, there is a toolbar with icons for pinning, printing, refreshing, and deleting, followed by an 'SQL' label. To the right of the toolbar, it says 'All Rows Fetched: 4 in 0.019 seconds'. Below the toolbar is a table with three columns: 'COUNTRY_ID', 'COUNTRY_NAME', and 'REGION_NAME'. The table contains four rows of data, ordered by country name. The first row is for Canada (CA), the second for Germany (DE), the third for the United Kingdom (UK), and the fourth for the United States of America (US).

	COUNTRY_ID	COUNTRY_NAME	REGION_NAME
1	CA	Canada	Americas
2	DE	Germany	Europe
3	UK	United Kingdom	Europe
4	US	United States of America	Americas